CHAPTER 20 · DERIVATIONS

OSCILLATIONS

3 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 VELOCITY IN SHM

SYLLABUS 20.1(C)

RESULT $v = \pm\omega\sqrt{A^2 - x^2}$

1. $x = A\sin\omega t$, so $v = \frac{dx}{dt} = A\omega\cos\omega t$.
2. $\cos\omega t = \pm\sqrt{1 - \sin^2\omega t} = \pm\sqrt{1 - \frac{x^2}{A^2}}$.
3. $\mathbf{v = \pm\omega\sqrt{A^2 - x^2}}$; maximum ωA at $x = 0$, zero at $x = \pm A$.

2 KINETIC AND POTENTIAL ENERGY IN SHM

SYLLABUS 20.2(A)

RESULT $K = \frac{1}{2}m\omega^2(A^2 - x^2)$, $U = \frac{1}{2}m\omega^2x^2$

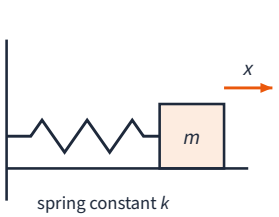
1. $K = \frac{1}{2}mv^2 = \mathbf{\frac{1}{2}m\omega^2(A^2 - x^2)}$.
2. Restoring force $F = -m\omega^2x$. Work done against it from 0 to x: $U = \int m\omega^2x\,dx = \mathbf{\frac{1}{2}m\omega^2x^2}$.
3. Total energy $K + U = \frac{1}{2}m\omega^2A^2$: constant.

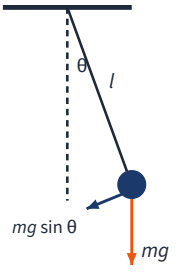
3 PERIOD OF A SPRING-MASS SYSTEM AND A SIMPLE PENDULUM

SYLLABUS 20.3(A)

RESULT $T = 2\pi\sqrt{\frac{m}{k}}$, $T = 2\pi\sqrt{\frac{l}{g}}$

1. **Spring–mass**: displaced x from equilibrium, restoring force $F = -kx$. $ma = -kx$, so $a = -\frac{k}{m}x$.
2. This is SHM ($a = -\omega^2x$) with $\omega^2 = \frac{k}{m}$: $\mathbf{T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}}$.
3. **Simple pendulum**: displaced by angle θ, restoring force along the arc $= -mg\sin\theta \approx -mg\theta$ for small θ, and $\theta = \frac{x}{l}$.
4. $ma = -\frac{mgx}{l}$, so $a = -\frac{g}{l}x$: $\omega^2 = \frac{g}{l}$ and $\mathbf{T = 2\pi\sqrt{\frac{l}{g}}}$.





Valid for small oscillations (θ less than about 10°).